

Credit card companies are making transaction technology more and more secure.

With the introduction of chip-insert methodologies for buying things at stores (collaboratively invented by both Roland Moreno and Europay-Mastercard-Visa) a world has been created, that subverts many of the threats of yesterday. In 2014, the world was even more unsafe for buyers and sellers than it is now, and phasing out swipe card technology since then, has been advantageous.

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Let's discuss what it is exactly that chip cards do:

First, the terminal powers the chip through tiny metal contacts on the card.

Authentication is where the magic happens. Your card's chip has a mini-computer that sends of a cryptogram to your bank that represents the transaction, which refreshes itself for every new transaction.

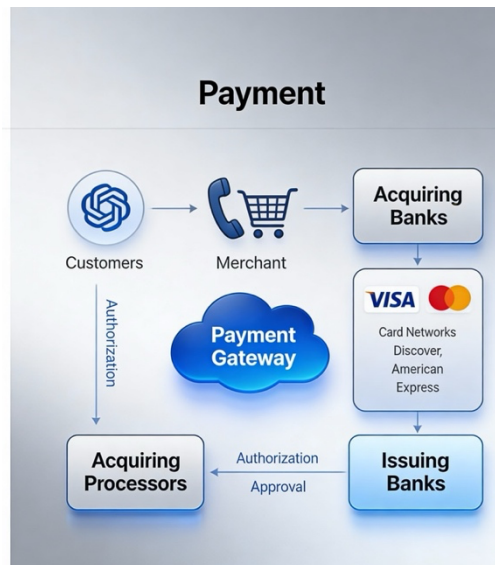
The chip card uses digital certificates and cryptographic keys, which alert the payment terminal that the card hasn't been tampered with or cloned. Dynamic data generation is at play here, updating information in real time.

Verification and approval happen within seconds. The cryptogram and other information about the purchase are sent to the bank. The bank uses its saved user account data to match everything up to your location, point of purchase, and other spender habits that may be valuable knowledge to have in quickly determining if there's a fraudulent transaction being undertaken in live time.

Banks these days use machine learning to build a behavioral profile of your spending, and if a spending alert deviates from it, they know that something's up.

Verification can also sometimes include digital body language including how you move your mouse or your typing speed on a computer. New or unknown IP addresses used for the transaction work the same way, alerting your bank that something is up.

The bank then either passively monitors the transaction, triggers step-up authentication, or hard blocks the card if something *really* suspicious is taking place in the realm of credit card info theft.



Roland Moreno was a French inventor from the 1970s creating the first integrated circuit card then. The spread of the card's initial design was prevalent in the early 1980s but only in France.

Things are getting safer for people with big bank accounts who want to protect where their money is going as well as who has access to their buying power. Families don't often share credit card accounts, but they can. Middle-class folk have identity threats to worry about too. Someone trying to steal your money is serious business.

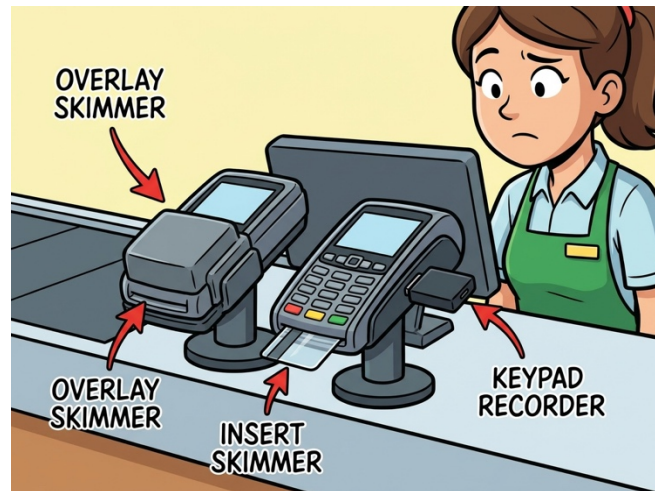
Types of Skimming:

- Magnetic-stripe skimming
- Point of sale employee skimming (inside job)
- Shimmying (related to chip cards) aka chip skimming
- Near Field Communication Skimming
- Online skimming

Magnetic-stripe skimming involves adding a device to a cash wrap machine at a gas station or vendor's machine, which can copy the data that is presented to it via swipe. It usually looks bulky, can be shaken off, LED lights on the machine might be misaligned. The plastic card slot will have a thicker mouth. There are also cameras that can be installed on places of transaction—particularly gas stations—that can watch you as you type your pin in or potentially scan your face for use for facial ID scans (another form of identity theft). Skimmers also use fake keypads (pin pad overlays) installed over preexisting keypads to save keystroke data through a sensitive circuit board grid that exists on the counterfeit.

Shimming involves bringing a little scanner near your pocket, which can hold tap transactional data, that is also stored on your chip. Shimming can also entail a paper thin shim is inserted into the card slot that can read communication between the chip and terminal. These technologies

(thankfully) are harder to install and less effective. So EMV upgrading their technologies paid off in resultant securer transactions.



An employee can insert a skimmer to scan through all the credit card data that was transferred to the machine throughout the day.

Joints that only use swipe are still privy to this sort of threat, which includes things like skimming and copying transaction information to complete further transactions. Thankfully in 2025, it's hard to find a vendor that only does swipe. They even have wireless cashiers that complete electronic transactions for all without a physical cash wrap machine that sits at the point of purchase desk.

Near field communication skimming involves an attacker carrying a portable NFC reader or modified phone. The attacker gets very close to your pocket, bag, or wallet, then the reader activates your card and captures transmitted data. They do not transfer CVV or pin, but this kind of skimming can still result in cloning, tracking, or online fraud attempts. This kind of skimming is more popular in crowded places and tourist areas. It is more likely to occur if you have a poorly shielded wallet. A lot of data can be snatched from you because of near field communication skimming such as card ID and balance.



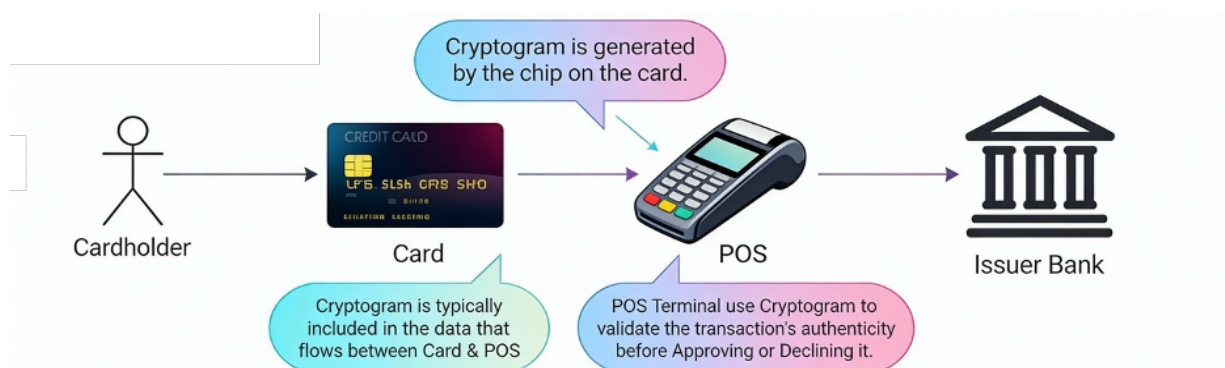
RFID chips on passports (e-passport chips) rely on the same radio technology that EMV invented in 2014 for digital transactions at point of purchase. These can be skimmed by NFC skimming too. Both chips employ the same temper resistant secure chip with encrypted memory, hardware crypto engines, and probing and glitching. Contained within the two is very similar secure element hardware. EMV's technology was launched at around the same time, so in a way EMV contributed to development of this kind of technology.



Credit card chip technology is new to the world, in terms of human civilization and the dawn of modern human technology from the 1800s industrial revolution. As we move in our Model T's towards telephones in our pockets, it's important to remember Bill Gate's wise words: "The advance of technology is based on making it fit in so that you don't really even notice it, so it's part of everyday life."

It's important, not to take for granted the implications of this safer technology. As identity theft becomes more common, and banks and thieves try to take advantage of the common gullible pedestrian, let's take every precaution not to get pick pocketed and cause a bunch of unnecessary overhead in tracking down theft. Imagine you spend \$200 a week, and somebody rings your account up for a \$2000 jewelry purchase. Are you just going to let them steal the pearls?

Or will you intervene and make sure the theft doesn't proceed as it intended to. Think about how the business in the jewelry industry has to track down the thief, rescind a shipped product, and possibly lose important merchandise that has recently fraudulently been purchased.



Let's talk about **cryptograms**.

Cryptograms are one-time codes sent by the credit card chip which are almost impossible to skim and contain a one-time-use transactional digit that carries and encrypts the details of your purchase. The cryptogram is signed using secret keys and the bank verifies it in real time. Each transaction is mathematically unique, making card data reuse useless. This can include demographic information and information that is given at the point of sale like your social security number, bank verification details, your email address, and place of living (zip code). The cryptogram also obviously contains the place where you're purchasing a good or service like a retail store (711) or a car rental place like [Hertz](#). The reason identity theft is so rampant, is because people were able to use your card swipes for data with the use of skimming technologies.



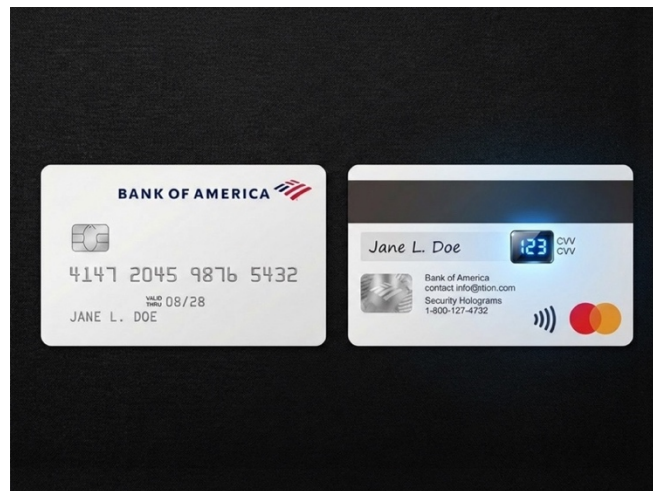
There is a plastic contraption you can connect to gas station cashier service pumps, with their card readers and interactive screens, that can skim cards that are entered into it for use. The way they work is: they store the swipe's codes and keys. The information stored in a credit card swipe barcode wasn't unique back in the day. The code left behind by a credit card as an impression on the cashier device at the vendor or outlet, existed only once (static), and so it was easier to steal data from it, perform identity theft, or buy a bunch of things that you don't want on your card.

A way to make this even more secure these days, is a daily corruptible CVV code file printed upon glass on a tiny LED built into the thickness of your credit card. These are a secret agent's way of saying "Stay the heck out of my space."

The CIA definitely had something to do with the sophistication of crime being committed in the world of person to person cyber warfare, and assaults conducted through the wires of social communication platforms through potential data theft. The CIA also had something to do with the imperative that exists for technology like this to be in use. Commonly, an agency of this power over Pentagon operations and economic health of the country, and the president's take on the army, I think the EMV was prompted by them, RAW, and military intelligence 6 to upgrade

this technology as soon as possible, with the written approval of their stylish heavy ball point pen.

Imagine if scanning your card was individualized by you planned daily to be a part of your renewal in the world of code safety. This is important data to protect. Protecting it in your pocket is the right way to go. Demographic data collected at the point of purchase is yours to give away, but just be safe.



Cryptographic keys and gas-station skimmers were mentioned previously in the article, but to really understand what's going on with the technology here, you need to try and wrap your mind around the lightning speed directives performed by the use of something like WIFI, Bluetooth, and radio wave signals (near field communication). Again, an interaction of just your card and the bank, communicating with each other in real time, but in seconds, to verify a transaction. Should your card be used 100s of miles away from home, the bank would instantly alert you via text, and the card could be locked, so that more purchases aren't made in your name. Fraudulent purchases are typically prevented through the payment gateway's newly created security systems

though. Acquiring processors take their time with authorizing a purchase's happenstance. They collect authorization from customers and acquire authorization from your bank as well.

The same way a computer can get attacked by a virus via an internet portal, so too can opportunities present themselves to you in the form of credit card data theft. This backlogs the system. You have to hear me out here. Imagine spending a million dollars of fraudulent material purchase through a brand like Burberry. What kind of idea would that send the brand?

What could they do about it?

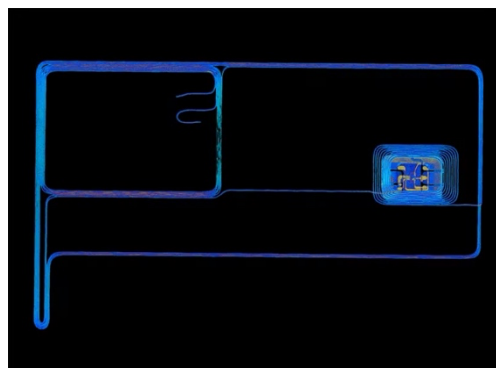
The stolen merchandise, then becoming more common, could cause all kinds of havoc to movie stars, conmen, and burglars alike.

Europay is a payment method that is universal in the EU. Mastercard primarily operates in the United States. Visa is a traveler's credit card and a flying transmission travelling plan all rolled into one. Holding down your identity in a new town and travelling frequently with a lavish lifestyle requires a safe visa profile.

Authentication and verification extends past just the bounds of a chip card inserted to a terminal. When you're making an online transaction at a trusted vendor that has your account data, there is metadata stored by the vendor's website, which will actually make the credit card transaction more secure by monitoring mouse movements around the screen, their speed, and the stylistic transitions of arches and diagonals. If the movement of the mouse is suspicious, the bank will block the transaction.

Chip security is getting stronger, as newer versions of the chips are installed on cards, that ensure better safety. They make data processing faster and shield against outside threats, who may try and learn your data, well. As the technology evolves it's important to understand it's advantages and positive outcomes.

The electro circuitry of a card's chip is sophisticated, intricate, efficient, and powerful. While the complexities of the innerworkings of silicon science can't be explained in tandem to the mechanics of circuit initialization, there is much to be understood about the circuitry of a credit card chip, that anyone can understand given the right knowledge and information, to know what kind of technology is stored in your pocket.



Those 6–8 gold rectangles visible in your chip without a magnifying glass and with the naked eye are standardized electrical contacts. You can see 3 gold rectangles carved into the plastic on either side of an array of pads :

Pad	Purpose
VCC	Power ($\approx$ 5V or 3V from the terminal)
GND	Ground
CLK	Clock signal
RST	Reset
I/O	Serial data in/out
RFU	Reserved / vendor-specific

On top of this architecture which presents itself to you when you zoom in on the chip, there is a bonded silicon die.

The silicon die has a microcontroller unit (MCU). It integrates CPU, Memory, Input/Output peripherals, timers, clocks, and communication interfaces.

Here's a little bit about how they're packaged. There's wafer fabrication, or circuits built onto the silicon wafer which forms the silicon die chip. In this case, we're looking at a part of a greater chip, so a smaller chip within the bigger chip. Zooming in on some dicing occurring within the wafer. Each sub-die is mounted and electrically connected. It's all packaged in a form that includes encapsulation (like a box with bubble wrap inside for your expensive mug) or padding if you're familiar with the coding term with external leads added to expand and branch out for connectivity (much like neurons and their dendrites in your brain at points of synapse). External leads can also be a way to ensure connection to nearby chip parts, some of which were described earlier.

The silicon die part of the chip also has multiple memory methods and uses ROM, Flash, and RAM. Flash technologies have advanced since their origins in 1980s, the provenance decade of cult film classic *Back to the Future*. The ROM holds the immutable OS and core crypto routines. Flash holds the card number, keys, limits, and counters. RAM then is responsible for temporary transaction data operations, or the exchange of information between two devices. The silicon die is the material the chip is made of. The integrated circuit is fabricated here, but it's not located in the center of the chip.

The chip has 4 layers on top of the silicon die, and the silicon die surface area exists somewhere in the middle of the 3 layers of the Earth. You can think of the crust, mantle, and core metaphorically as we journey inwards. The first layer is the card plastic body at the top, then an adhesive layer, chip module substrate, wire bonds / bumps, and finally the silicon die wafer. There are then 3 other layers that sandwich below the silicon die. The encapsulation epoxy lies directly below, and then gold contact pads, and finally: card reader pins. Think about these words and let them dawn on you information you can utilize as a step before understanding modern nano-technologies.



Let's understand cryptographic hardware. Circuitry occurring here is of great interest to us today. There are accelerators. 4 of them.

- AES (Advanced Encryption Standard)
- 3DES (Triple Data Encryption Standard)
- RSA (Rivest–Shamir–Adleman)
- ECC (Elliptic Curve Cryptography), and secure random number generation (TRNG).

If you've ever used a key generator to install stolen software through file sharing, you know all about TRNG technologies.

During a transaction, the all surrounding cryptographic hardware, or the internal infrastructure needed to power your chip, the form of your chip take care of transactions, one at a time. The terminal, upon connecting, sends a challenge and then the chip generates a unique cryptogram and the cryptogram is signed using secret keys, including the pin you enter into the terminal verifying the debit transaction. The bank then has technologies in place on their servers and software capabilities to verify the transaction in real time, assuring security protocol.

There's no battery in the card. Circuitry is booted up from a **jolt in the terminal**. However it's important to note that electromagnetic conduction is also at play here through tap and NFC (Near Field Communication). NFC allows the chip's tap capabilities to communicate with the mother board of the credit card. It boots instantly, runs for milliseconds, then shuts down when you press eject, or pull out the card.

NFC circuitry is complicated stuff. Let's take it step by step as we move towards understanding it. There is a copper antenna loop embedded into the plastic. The wire emits radio frequencies creating a 13.56 MHz field. The antenna harvests energy, inhabiting an electrical field. The chip then communicates wirelessly, sending data messages. Again, only turning on for a second. Time must be figurative in the non-thinking heads of these technologies. Imagine WIFI or bluetooth waves thinking for themselves and re-directing their signals towards you.

The card has a lot of abilities to take countermeasures for security purposes. We're done talking about the parts of the chip. This is the chip's capabilities in relation to security. In a way, this is a microcosm of chip technology in the world of cyber security. So pay attention.

The chip's circuitry includes defenses like tamper-detect layers that contain voltage, clock capabilities, and temperature storage. The chip will melt if it overheats. While it has no cooling mechanisms, it is in its ability to save itself, should it be presented by too much heat, by warning users through its master OS. Randomized execution timing, then presents skimmers with a factor of confusion, so that the signals aren't disrupted in the process of completing your transaction securely. There's also a shield of protective layering material that goes over the silicon. The chip needs durability and so its layers assure efficiency and robust hardware. There is automatic memory wipe capability for security purposes too. If someone tries probing the chip electrically, the chip can self-destruct its secrets. This chip technology is advanced for being able to end its short life as a fruit fly in your house. The fly is being told to save itself in flight from a slammed human hand. Hardware-level defenses are efficient in protecting the card's data from fraudulent action taken to steal from you.

The clock signal connects to the WIFI to ascertain the time, even with accordance to daylight savings, once prompted to do so, with the help of terminal technologies as well. They race to connect you to your bank to authorize the transaction and receive your goods. A working digital clock is the least these puppies could hold.

Why EMV Chips Beat Magnetic Stripes

Magnetic Stripe	EMV Chip
Static data	Dynamic cryptograms
Easy to clone	Cryptographically unique
No CPU	Secure microcontroller
Offline	Real-time authentication

This whole system is governed by EMV which defines how the circuitry, protocols, and crypto all work together.

The terminal zaps the card with 5V and off it goes. The chip spurts into action.

It zips up your personal data, shopper imprint, and buying power, and sends you off into a world of banks engulfed by good will, doing their best to help you in their transactions of money.

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Blue and gray payment diagram & gradient pink blue turquoise diagram:

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